

Statistics Tutorial 3

Problem 1

1. A company conducted a customer satisfaction survey across three regions (North, South, and West) to determine if customer satisfaction is independent of the region. The responses were categorized as "Satisfied," "Neutral," and "Dissatisfied." The survey results are shown in the table below:

Region	Satisfied	Neutral	Dissatisfied
North	45	30	25
South	40	35	25
West	50	25	25

- (a) State the null and alternative hypotheses for this test.
- (b) Calculate the expected frequencies for each cell in the table under the assumption that satisfaction is independent of region.
- (c) Compute the Chi-Square test statistic using the formula:

$$\chi^2 = \sum \frac{(O_{i,j} - E_{i,j})^2}{E_{i,j}}$$

where $O_{i,j}$ is the observed frequency and $E_{i,j}$ is the expected frequency.

- (d) Determine the degrees of freedom for this test.
- (e) At a significance level of $\alpha = 0.05$, should you reject or fail to reject the null hypothesis? Use a Chi-Square distribution table or software to find the critical value or p-value.
(Hint: Critical Value - If test statistic > critical value: Reject H_0 ; If test statistic < critical value: Fail to reject H_0 . P-value - If p-value < α : Reject H_0 ; If p-value > α : Fail to reject H_0)

1. We're testing whether there's a relationship between two categorical variables: region (North, South, West) and satisfaction level (Satisfied, Neutral, Dissatisfied).

Step 1: Hypotheses

- **Null Hypothesis** (H_0): Customer satisfaction is independent of region
- **Alternative Hypothesis** (H_1): Customer satisfaction is dependent on region

Step 2: Calculate Row and Column Totals

Region	Satisfied	Neutral	Dissatisfied	Row Total
North	45	30	25	100
South	40	35	25	100
West	50	25	25	100
Column Total	135	90	75	300

Table 1: Observed Frequencies with Row and Column Totals

Step 3: Calculate Expected Frequencies

Expected frequency formula:

$$E_{i,j} = \frac{\text{Row Total} \times \text{Column Total}}{\text{Grand Total}}$$

For each cell:

$$\text{North-Satisfied: } E_{11} = \frac{100 \times 135}{300} = 45$$

$$\text{North-Neutral: } E_{12} = \frac{100 \times 90}{300} = 30$$

$$\text{North-Dissatisfied: } E_{13} = \frac{100 \times 75}{300} = 25$$

Region	Satisfied	Neutral	Dissatisfied
North	45	30	25
South	40	30	25
West	45	30	25

Table 2: Expected Frequencies

Step 4: Calculate Chi-Square Statistic

$$\chi^2 = \sum \frac{(O_{i,j} - E_{i,j})^2}{E_{i,j}}$$

Calculations for each cell:

$$\text{North-Satisfied: } \frac{(45 - 45)^2}{45} = 0$$

$$\text{South-Satisfied: } \frac{(40 - 45)^2}{45} = 0.556$$

$$\text{West-Satisfied: } \frac{(50 - 45)^2}{45} = 0.556$$

Total Chi-Square statistic = 2.778

Step 5: Degrees of Freedom

$$df = (rows - 1)(columns - 1) = (3 - 1)(3 - 1) = 4$$

Step 6: Decision Making

At $\alpha = 0.05$:

- Critical value for $df = 4$ is 9.488
- Calculated $\chi^2(2.778) < \text{critical value } (9.488)$
- p-value = 0.595 > 0.05

Decision: Fail to reject the null hypothesis.

There is not enough evidence to conclude that customer satisfaction depends on region.

Reference - Using Chi-Square Distribution Table:

- (a) Critical value - At $\alpha = 0.05$ significance level with $df = 4$, looking at the chi-square table: The critical value is 9.488.
- (b) Approximate p-value - Find row with $df = 4$, look for values that our $\chi^2(2.778)$ falls between. Our 2.778 falls somewhere between $\chi^2(0.20) = 5.989$ and $\chi^2(0.90) = 1.064$. Therefore p-value is between 0.20 and 0.90.

Problem 2

In an experiment, asphalt with low levels of air voids(2-4%), with medium level of air voids (4-6 %) and high(6-8 %) are tested.

Refer to table 1 and answer the following:

- (a) Do the different levels of air voids significantly affect the mean retained strength. Use $\alpha = 0.01$
- (b) Find the p value of the F statistic in part (a)

Table 1: Strength of asphalt at different concentrations

Low	Medium	High
106	80	78
90	69	80
103	94	62
90	91	69
79	70	76
88	83	85
92	87	69
95	83	85

$$\bar{X}_{low} = \frac{\Sigma x_{low}}{8} = 92.875$$

$$\bar{X}_{medium} = \frac{\Sigma x_{medium}}{8} = 82.125$$

$$\bar{X}_{high} = \frac{\Sigma x_{high}}{8} = 75.5$$

$$\bar{X} = 83.5$$

Formulas for ANOVA

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F-Stat
Between Treatments	$n \sum_{i=1}^a (\bar{y}_{i.} - \bar{\bar{y}}_{..})^2$ or SSB	a-1	MSB = SSB/DoF	F = MSB/MSE
Error within treatments	$\sum_{i=1}^a \sum_{j=1}^n (y_{i,j} - \bar{y}_{i.})^2$ or SSE	N-a	MSE = SSE/DoF	
Total	$\sum_{i=1}^a \sum_{j=1}^n (y_{i,j} - \bar{\bar{y}}_{..})^2$ or SST	N-1	MST = SST/DoF	

Compare F calculated against the F-distribution with a-1,N-a degrees of freedom and get a p-value

The null hypothesis for the given problem is :

$$H_0 : \mu_L = \mu_M = \mu_H$$

The alternate hypothesis is given by: H_1 : Atleast 1 of the means is not equal between groups= $k-1=3-1=2$ degrees of freedom

degrees of freedom within groups = $n-k=24-3=21$

total degrees of freedom= $n-1=23$

$$\bar{X}_{low} = \frac{\Sigma x_{low}}{8} = 92.875$$

$$\bar{X}_{medium} = \frac{\Sigma x_{medium}}{8} = 82.125$$

$$\bar{X}_{high} = \frac{\Sigma x_{high}}{8} = 75.5$$

$$\bar{X} = 83.5$$

(This is the overall mean).

$$SS_t = \Sigma(x_i - \bar{X})^2 = 1216 + 584 + 986 = 2786$$

$$SS_b = n * \Sigma(\bar{X}_i - \bar{X})^2 = 1230.25$$

$$SS_w = SS_t - SS_b = 1555.75$$

$$MSB = \frac{SS_b}{df_b} = 1230.25/2 = 615.125$$

$$MSE = \frac{SS_w}{df_w} = 1555.75/21 = 74.083$$

Let us now find the F-value.

$$F = \frac{MSB}{MSE} = 615.125/74.0833 = 8.303$$

From F tables for

$$\alpha = 0.01$$

,

$$f_c = F_{2,21} = 5.78$$

. As

$$F > f_c$$

, the differences are statistically significant and we reject the null hypothesis.

p-value calculation:

$$df_1 = 2, df_2 = 21, f = 8.303$$

,from f table for probability, we get:

$$p(f > 8.303) = 0.0022$$